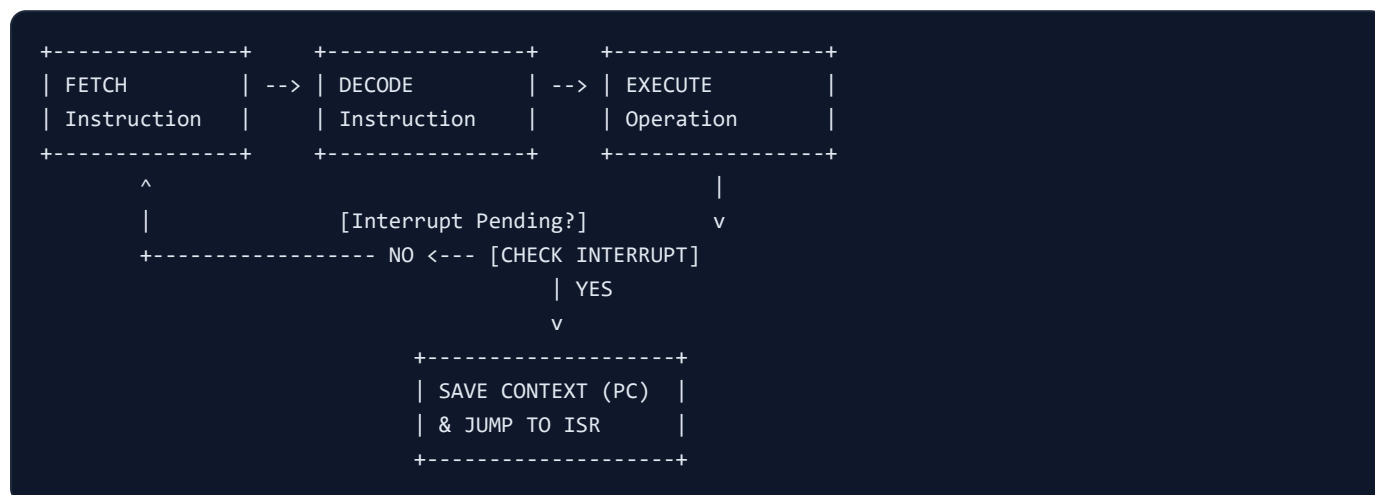


3-Mark Questions & Answers — MPCA Unit 2: Processor Organization & Architecture

Q1. Explain the basic instruction cycle with interrupt processing.



- **Fetch:** Load instruction from memory (**PC** -> **MAR** -> **RAM** -> **IR**).
- **Decode:** Decode opcode and fetch required operands.
- **Execute:** Perform operation in ALU.
- **Interrupt Check:** Check if a hardware/software interrupt flag is set; if yes, push context to stack and jump to Interrupt Service Routine (ISR).

Q2. Differentiate between hardwired and microprogrammed control unit.

Feature	Hardwired Control Unit	Microprogrammed Control Unit
Implementation	Fixed logic gates, flip-flops, decoders.	Microcode software stored in Control Memory (ROM).
Speed	Very Fast (hardware gate speeds).	Slower (requires ROM access cycles).
Flexibility / Modifiability	Rigid; hard to modify or update.	Flexible; easy to update microcode.
Cost & Design	Complex design, costly for complex ISA.	Systematic design, cheaper for CISC.

Q3. Explain types of pipeline hazards (structural, data, control) briefly.

1. **Structural Hazard (Resource Conflict):** Occurs when two instructions in different stages attempt to access the same physical hardware resource simultaneously (e.g., single memory port for instruction fetch and data read).
2. **Data Hazard (Data Dependency):** Occurs when an instruction depends on the result of a previous instruction that has not yet completed execution (RAW, WAR, WAW hazards).
3. **Control Hazard (Branch Hazard):** Occurs due to conditional or unconditional branch instructions that alter the sequential flow of the Program Counter (**PC**).

Q4. Explain Flynn's classification (SISD, SIMD, MISD, MIMD) with examples.

1. **SISD (Single Instruction, Single Data):** Classic single-core processor executing one stream of scalar data (e.g., Intel 8086, traditional single-core CPUs).
 2. **SIMD (Single Instruction, Multiple Data):** One instruction operates on vector/matrix array elements concurrently (e.g., GPUs, Intel AVX/SSE).
 3. **MISD (Multiple Instruction, Single Data):** Multiple processors execute different instructions on the same data stream (e.g., Space Shuttle fault-tolerant flight control).
 4. **MIMD (Multiple Instruction, Multiple Data):** Independent processors executing different instructions on independent data streams (e.g., Intel Core i9 multi-core CPUs, supercomputers).
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Q5. What are micro-operations? Give examples.

Micro-operations are elementary functional operations executed by CPU hardware during a single clock pulse to perform register transfers, ALU calculations, or bus reads.

- **Examples:**

1. **MAR ← PC** (Transfer Program Counter to Memory Address Register)
2. **DR ← M[MAR]** (Fetch memory contents into Data Register)
3. **PC ← PC + 1** (Increment Program Counter)
4. **AC ← AC + DR** (Add Data Register to Accumulator)